4 Chapter Four: REGRESSION AND CORRELATIONS

4.1.1 LINEAR REGRESSION

When two varieties x, y, whose measurements have been shown on a graph and look like the following figures below:

It is natural to try to describe the linear, or approximately linear, relation between them by an equation, the **linear regression** equation. This expresses one variate, the **dependent** variable *y*, in terms of the other, the **independent** variable *x*.

We can properly write $y = b_0 + b_1 x$ to show how *y* alters when *x* is altered, where "b_0" is the **intercept**, or that value taken by *y* when *x* = 0, while "b_1" is the **slope** or gradient of the line, the average increase in *y* for unit increase in *x*. We can choose, and fix, the values of *x* to use in an experiment designed to study this relation, but we have no control over the resulting *y*.

4.1.2 ORDINARY LEAST SQUARES METHOD (OLS)

Is a technique of fitting the "best" curve to a set of observations $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$. If the points are plotted on a rectangular co-ordinate system, it is possible to visualize the relationship between Y and X. The plotted set of points is called a **scatter diagram**.

4.2 Example

The process of estimating the parameters of the equation y = f (x) using the method of least squares is referred to as **regression**.

4.2.1 Simple Linear Regression Analysis

Given the data $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$, the problem of linear regression analysis is to find the estimates of "a" and "b" in the equation $y = a + bx$

which best fit the data. The line $\hat{y} = \hat{a} + \hat{b}x$ (where $\hat{a}$ and $\hat{b}$ are the least squares estimates of a and b respectively) is called the least squares regression line of Y on X.

4.2.1.1 Note

We assume that the values of the independent variable (i.e. X) are obtained without error. The predicted value of *y* at a given value of *x* is given by $\hat{y} = \hat{a} + \hat{b}x$.

In general, $\hat{y}_i \neq y_i$.

Define $d_i = y_i - y_i^{(p)}$ (is the difference between the observed value and the predicted value)

The approximating line L is the best fitting line (or the least squares line if $\sum_{i=1}^{n} d_i^2$ is a minimum)

4.3 Method: To find the least squares regression line of y on x

Given the data $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$, we minimize:

$$S = \sum_{i=1}^{n} d_i^2$$

$$= \sum_{i=1}^{n} \left(y_i - y_i^{(p)}\right)^2 = \sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]^2 = \sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_i\right)^2$$

The least squares estimates of b_0 and b_1 satisfy:

$$\frac{\partial S}{\partial b_0} = 0 \quad and \quad \frac{\partial S}{\partial b_1} = 0 \;\} \; for\ a\ \min imum$$

$$\frac{\partial S}{\partial b_0} = -2\sum \left(y_i - b_0 - b_1 x_i\right) = 0 \quad \cdots\cdots\cdots\cdots (i)$$

$$\frac{\partial S}{\partial b_1} = -2\sum \left(y_i - b_0 - b_1 x_i\right) x_i = 0 \quad \cdots\cdots\cdots\cdots (ii)$$

Solving equations (i) and (ii) simultaneously, we get

$$\hat{b}_1 = \frac{n\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2 - \left(\sum x_i\right)^2} \quad \text{and} \quad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

The line $\hat{y} = \hat{b}_0 + \hat{b}_1 x$ is called the least squares regression line of Y on X since it is estimated from X.

4.4 **Example**

For the given data below, calculate the least squares regression line of Y on X. Estimate the value of Y when X = 12, X = 15.

Mean Temp. (X)°C	7	5	5	9	10	15	18	19
Energy Consumption (Y)	29	30	34	27	23	25	19	18

4.5 **Solution**

x	**y**	**xy**	**x^2**
7	29	203	49
5	30	150	25
5	34	170	25
9	27	243	81
10	23	230	100
15	25	375	225
18	19	342	324
19	18	342	361
$\sum x = 88$	$\sum y = 205$	$\sum xy = 2055$	$\sum x^2 = 1190$

$$\hat{b}_1 = \frac{n\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2 - \left(\sum x_i\right)^2}$$

$$= \frac{(8)(2055) - (88)(205)}{(8)(1190) - 88^2} = -0.9$$

$$\hat{b}_0 = \bar{y} - \hat{b}_1\bar{x} = \frac{205}{8} - (-0.9)(88/8) = 35.525$$

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x = 35.525 - 0.9x$$

The predicted energy consumption when the temperature is 12^0C is given by:

$$\hat{y} = 35.525 - 0.9x = 35.525 - 0.9(12) = 24.73 \; units$$

When x = 15^0C; $\hat{y} = 35.525 - 0.9x = 35.525 - 0.9(15) = 22.03 \; units$

Note the actual value observed is 25 units, $\therefore d = 25 - 22.03 = 2.97$

4.5.1 Simple Linear Regression Model

We have a set of data $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$ where it is assumed that for fixed X; the conditional density of Y given X is normally distributed with $Mean = b_0 + b_1 x$ and $Variance = \sigma^2$, where b_0, b_1, and σ^2 are unknown constants.

i.e. $Y/X \sim N(b_0 + b_1 x, \sigma^2)$

$E(Y/X) = b_0 + b_1 x$

The linear relationship between Y and X can be made stochastic by adding an error term μ. That is; $y = b_0 + b_1 x + \mu$. This is called a Simple Linear Regression Model. Normal linear regression analysis is concerned with the:

(i) Estimation of the parameters b_0, b_1, and σ^2 (if σ is unknown).

(ii) Testing of hypothesis concerning these parameters.

(iii) Prediction based on the estimated regression model.

4.6 Classical assumptions of linear regression model

(i) $\mu \sim N(0, \sigma^2)$

(ii) X can be measured without error, i.e. $E\left(Y/X\right)=b_0+b_1x$ or

$Y/X \sim N\left(b_0+b_1x,\sigma^2\right)$

Note that these assumptions are necessary for testing hypothesis but <u>not</u> for obtaining the estimates of b_0 and b_1.

We have seen earlier that the least squares estimates of b_0 and b_1 are given by:

$$\hat{b}_1=\frac{n\sum x_iy_i-\left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2-\left(\sum x_i\right)^2} \quad \text{and} \quad \hat{b}_0=\bar{y}-\hat{b}_1\bar{x}$$

The estimated regression equation is given by $\hat{y}=\hat{b}_0+\hat{b}_1x$. $\hat{y}$ is called the predicted value of Y corresponding to a given value of X.

$\hat{b}_1$ can also be expressed in each of the following alternative forms:

$$(i)\ \hat{b}_1=\frac{\sum\left(x_i-\bar{x}\right)\left(y_i-\bar{y}\right)}{\sum\left(x_i-\bar{x}\right)^2} \quad (ii)\ \hat{b}_1=\frac{\sum\left(x_i-\bar{x}\right)y_i}{\sum\left(x_i-\bar{x}\right)^2} \quad (iii)\ \hat{b}_1=\frac{\sum\left(y_i-\bar{y}\right)x_i}{\sum\left(x_i-\bar{x}\right)^2}$$

4.6.1 Test of Significance of the parameter estimates

To test for the significance of b_0 and b_1, we require the mean and variance of $\hat{b}_0$ *and* $\hat{b}_1$.

Model: $y_i=b_0+b_1x_i+\mu_i$ *where* $\mu_i\sim N\left(0,\sigma^2\right)$ $i=1,2,\cdots,n$

4.6.1.1.1 Assumptions

$(i)\ \mu_i\sim N\left(0,\sigma^2\right),\ i=1,2,\cdots,n$

$(ii)\ Cov\left(\mu_i,\mu_j\right)=0$ *for* $i\neq j$, *i.e. the errorterms are uncorrelated*

$(iii)\ x_i's$ are fixed known values.

4.6.1.2 Mean and Variance of $\hat{b}_0$ *and* $\hat{b}_1$

$$E(\hat{b}_1)=E\left[\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}\right]=b_1\frac{\sum(x_i-\bar{x})^2}{\sum(x_i-\bar{x})^2}=b_1$$

Therefore $\hat{b}_1$ is an unbiased estimate of b_1.

$$E(\hat{b}_0)=E(\bar{y}-\hat{b}_1\bar{x})=b_0+b_1\bar{x}+E(\bar{\mu})-\bar{x}E(\hat{b}_1)$$
$$=b_0+b_1\bar{x}+0-b_1\bar{x}=b_0$$

$$\therefore E(\hat{b}_0)=b_0$$

Therefore $\hat{b}_0$ is an unbiased estimate of b_0

4.6.1.3 To find variance of $\hat{b}_1$ (assuming σ^2 is known)

$$Var(\hat{b}_1)=Var\left[\frac{\sum(x_i-\bar{x})y_i}{\sum(x_i-\bar{x})^2}\right]=\frac{\sigma^2}{\sum(x_i-\bar{x})^2}$$

Note: $\hat{b}_0$ *and* $\hat{b}_1$ are normally distributed random variables because they are linear functions of normal variables $y_i^{'s}$. Therefore:

$$\hat{b}_1 \text{ is } N\left(b_1,\ \frac{\sigma^2}{\sum(x_i-\bar{x})^2}\right)$$

4.6.1.4 To find Variance of $\hat{b}_0$ (assuming σ^2 is known)

$$Var(\hat{b}_0)=Var(\bar{y}-\hat{b}_1\bar{x})$$
$$=Var\left(b_0+b_1\bar{x}+\bar{\mu}-\hat{b}_1\bar{x}\right)$$

which upon reduction becomes

$$Var\left(\hat{b}_0\right)=\sigma^2\left[\frac{1}{n}+\frac{\bar{x}^2}{\sum(x_i-\bar{x})^2}\right]$$

i.e. $\hat{b}_0 \sim N\left(b_0, \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right]\right)$

Since σ^2 is usually unknown, $S^2 = \frac{\sum(y_i - \hat{y}_i)^2}{n-2}$ is used as an unbiased estimate of σ^2.

4.7 **Theorem**

The random variable $V = \frac{\sum(y_i - \hat{y}_i)^2}{\sigma^2}$ has a χ^2 - distribution with (n - 2) d.f.

4.7.1 To construct a T – test

Recall that if U *is* $N(0,1)$ and V *is* $\chi^2(r)$ and further that U and V are independent, then the random variable $T = \frac{U}{\sqrt{V/r}}$ has a students t – distribution with r – degrees of freedom.

In our problem;

$U = \frac{\hat{b}_1 - b_1}{\sqrt{\sigma^2 / \sum(x_i - \bar{x})^2}}$ *is* $N(0,1)$ and

$V = \frac{\sum(y_i - \hat{y}_i)^2}{\sigma^2} = \frac{(n-2)S^2}{\sigma^2}$ is distributed as $\chi^2(n-2)$

Assuming that U and V are independent,

$$T = \frac{U}{\sqrt{V/r}} = \frac{\dfrac{\hat{b}_1 - b_1}{\sqrt{\dfrac{\sigma^2}{\sum (x_i - \bar{x})^2}}}}{\sqrt{\dfrac{(n-2)S^2}{\sigma^2 (n-2)}}}$$

$$= \frac{\left(\hat{b}_1 - b_1\right)\sqrt{\sum (x_i - \bar{x})^2}}{S}$$ has a students t – distribution with (n - 2) d.f.

Note that $Var(\hat{b}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$. If σ^2 is unknown we replace it by

$S^2 = \frac{\sum (y_i - \hat{y}_i)^2}{n-2}$.

The estimated variance of $\hat{b}_1 = \frac{S^2}{\sum (x_i - \bar{x})^2} = S^2_{\hat{b}_1}$. $S^2_{\hat{b}_1}$ is an unbiased estimate

of $Var(\hat{b}_1)$. $S^2_{b_1} = \sqrt{\frac{S^2}{\sum (x_i - \bar{x})^2}} = \frac{S}{\sqrt{\sum (x_i - \bar{x})^2}}$ is called the standard error of

$\hat{b}_1$.

Therefore $T = \frac{\hat{b}_1 - b_1}{S_{\hat{b}_1}}$.

To test the hypothesis $\begin{matrix} H_0 : b_1 = B_1 \\ H_1 : b_1 \neq B_1 \end{matrix}$ V / S the test is to compute $T = \frac{\hat{b}_1 - B_1}{S_{\hat{b}_1}}$

which is distributed as $t_{(n-2)}$ when H_0 is true. Compare this value of T with the tabulated values of "t" with (n - 2) d.f. at α – level of significance.

If $|T| > t_{\alpha/2}$ (*tabulated*) reject H_0.

In particular if $B_1 = 0$ we say that b_1 is significant if H_0 is rejected.

To test for the hypothesis $\begin{matrix} H_0 : b_1 = B_1 \\ H_1 : b_1 > B_1 \end{matrix}$ V / S, reject H_0 if $T > t_\alpha$.